# Investigating the cosmological properties of the Stigma Q-torsion model

Sinah Legner,[1, 2, *] Will Barker,[3, †] and Will Handley[2, 4, ‡]

[1] *Astrophysics Group, Cavendish Laboratory, JJ Thomson Avenue, Cambridge CB3 0HE, UK*

[2] *Kavli Institute for Cosmology, Madingley Road, Cambridge CB3 0HA, UK*

[3] *Central European Institute for Cosmology and Fundamental Physics, Institute of Physics of the Czech Academy of Sciences, Na Slovance 1999/2, 182 00 Prague 8, Czechia*

[4] *Institute of Astronomy, Madingley Road, Cambridge CB3 0HA, UK*

Abstract text goes here.

## CONTENTS

I Introduction . . . . . 1
II Cosmology . . . . . 1
III Methodology . . . . . 2
IV blah . . . . . 2
V Results . . . . . 2
VI Discussion . . . . . 2
VII Conclusion . . . . . 2
References . . . . . 3

## I INTRODUCTION

***Poincaré Gauge Theory*** — The Poincaré Gauge Theory (PGT) of gravity provides a natural framework for introducing torsion and curvature as independent dynamical quantities in spacetime. Within the catalogue of ghost-free, tachyon-free, and power-counting renormalisable PGT models identified in [1, 2], the *Stigma* model represents a distinct branch characterised by a non-zero Q-torsion sector. The general PGT Lagrangian admits a rich parameter space, of which the Stigma model occupies a specific region defined by the coupling constants $(\varsigma, \upsilon_2)$.

$$\begin{aligned}\mathcal{L} =& \frac{3M_{\rm P}^2\upsilon_2}{4\varsigma}\Big[\frac{1}{2}\big(4\varsigma-\varpi^2\big)R - g^{\mu\nu}\partial_\mu\varpi\partial_\nu\varpi \\ &+ \varsigma\phi^2 - \varpi^2\phi^2 + \frac{4}{3}\sqrt{J_\mu J^\mu}\Big] - M_{\rm P}^2\Lambda + \mathcal{L}_{\rm M} \\ J_\mu \equiv& \varpi^3\partial_\mu(\phi/\varpi) - \partial_\mu\phi\end{aligned} \tag{1}$$

$$\begin{aligned}H^2 =& \frac{1}{\varpi^2-4\varsigma} \\ &\times\Bigg[-\frac{4H_0^2\varsigma}{\upsilon_2}\Big(\Omega_{{\rm r},0}a^{-4}+\Omega_{{\rm m},0}a^{-3}+\Omega_{\Lambda,0}\Big) \\ &+2H\varpi\dot\varpi+\dot\varpi^2+\frac{4\left(H\varsigma\left(\varpi^2-1\right)+\varsigma\varpi\dot\varpi\right)^2}{\varsigma-\varpi^2}\Bigg]\end{aligned} \tag{2}$$

$$\begin{aligned}\ddot\varpi =& \frac{1}{a^2\left(\varsigma-\varpi^2\right)\left(\varsigma+\left(4\varsigma^2-1\right)\varpi^2\right)} \\ &\times\Bigg[-3\varsigma^2a\dot a\dot\varpi+6\varsigma\left(1-2\varsigma^2\right)a\varpi^2\dot a\dot\varpi \\ &+3\left(4\varsigma^2-1\right)a\varpi^4\dot a\dot\varpi+\left(4\varsigma^2-1\right)\varpi^5\left(\dot a^2+a\ddot a\right) \\ &+2\varsigma\varpi^3\Big(\left(1-4\varsigma\right)\dot a^2+\left(1-2\varsigma(1+\varsigma)\right)a\ddot a\Big) \\ &+\varsigma^2\varpi\Big(3\dot a^2+a\left(-4\varsigma a\dot\varpi^2+\left(4\varsigma-1\right)\ddot a\right)\Big)\Bigg]\end{aligned} \tag{3}$$

## II COSMOLOGY

### A. Behavior at radiation-dominated epoch

We look at the behaviour of the field equations at the radiation dominated epoch, by looking only at a fluid-dominated epoch, i.e. $H = 2/(3(1+w_i)t)$, where $w_i$ is the equation of state parameter for the fluid. We find that from the field equation of $\varpi$, which we assume the

* sl2091@cam.ac.uk
† barker@fzu.cz
‡ wh260@cam.ac.uk

field $\varpi$ is a constant.

$$\begin{aligned}\left.\frac{\partial L}{\partial \varpi}\right|_{\varpi=\varpi_i} \propto\ & \frac{\varpi_i}{(w_i+1)^2(\varpi_i^2-\varsigma)^2\varsigma} \\ & \times\Bigg[(3w_i-1)(4\varsigma^2-1)\varpi_i^4 \\ & +\varsigma^2\Big(-7+4\varsigma+3w_i(4\varsigma-1) \\ & -2\varpi_i^2\varsigma\big(1+2(\varsigma-3)\varsigma \\ & +w_i(6\varsigma(1+\varsigma)-3)\big)\Big)\Bigg]\end{aligned} \tag{4}$$

Solving $\varpi^2$ for this equation reveals that

$$\begin{aligned}\varpi_i^2 =& \frac{\varsigma}{(3w_i-1)(4\varsigma^2-1)} \\ & \times\Big[1+2(\varsigma-3)\varsigma+w_i\Big(6\varsigma(1+\varsigma)-3\Big) \\ & \pm 2\sqrt{(\varsigma-1)^2\Big(2-6w_i+(\varsigma+3w_i\varsigma)^2\Big)}\Big]\end{aligned} \tag{5}$$

We can see that the solution contain poles at $w_i = 1/3$ and $\varsigma = 1/2$. However, upon further investigation we realised that the pole of $w_i = 1/3$ is a false-pole. Subsituting $w_i = 1/3$ at Eq. (4) we find that the equation reduces to

$$\left.\frac{\partial L}{\partial \varpi}\right|_{\varpi=\varpi_i}^{w_i=1/3} \propto -\frac{(\varpi_i^2-1)(\varsigma-1)\varpi_i\varsigma}{(\varpi_i^2-\varsigma)^2} \tag{6}$$

The numerator shows that at radiation-dominated epoch, $\varpi_i = 1$ and $\varsigma = 1$ are the only two solutions. To ensure that the field equation goes to zero, one of them must be set to be true.

### B. Power series

$$\begin{aligned}\varpi =& 1 \mp \frac{H_0\Omega_{\mathrm{m},0}\sqrt{4\varsigma-1}}{8\sqrt{-\upsilon_2\Omega_{\mathrm{r},0}\varsigma}}\tau \\ & -\frac{3H_0^2\Omega_{\mathrm{m},0}^2(4\varsigma-1)(2+\varsigma(3\varsigma-2))}{128\upsilon_2\Omega_{\mathrm{r},0}(\varsigma-1)\varsigma(8\varsigma^2+\varsigma-3)}\tau^2 \\ & \pm\frac{H_0^3\Omega_{\mathrm{m},0}^3}{1024(\varsigma-1)^2}\left(\frac{4\varsigma-1}{-\upsilon_2\varsigma\Omega_{\mathrm{r},0}}\right)^{3/2} \\ & \times\frac{\tau^3}{(8\varsigma^2+\varsigma-3)(20\varsigma^2+\varsigma-6)}\times\Big(192\varsigma^6 \\ & +380\varsigma^5-720\varsigma^4-225\varsigma^3+175\varsigma^2+72\varsigma-36\Big) \\ & +\mathcal{O}(\tau^4)\end{aligned} \tag{7}$$

$$\begin{aligned}a =& \pm\frac{2H_0\sqrt{\Omega_{\mathrm{r},0}\varsigma}}{\alpha\sqrt{\upsilon_2(1-4\varsigma)}}\tau-\frac{H_0^2\Omega_{\mathrm{m},0}}{4\alpha\upsilon_2}\tau^2 \\ & \pm\frac{3H_0^3\Omega_{\mathrm{m},0}^2\sqrt{\varsigma(4\varsigma-1)}(\varsigma(24\varsigma+5)-2)}{64\alpha(-\upsilon_2)^{3/2}\sqrt{\Omega_{\mathrm{r},0}}(\varsigma-1)(8\varsigma^2+\varsigma-3)}\tau^3 \\ & +\frac{H_0^4\Omega_{\mathrm{m},0}^3(4\varsigma-1)}{512\alpha\upsilon_2^2\Omega_{\mathrm{r},0}(\varsigma-1)^2} \\ & \times\frac{240\varsigma^5-1510\varsigma^4-246\varsigma^3+357\varsigma^2+31\varsigma-6}{(8\varsigma^2+\varsigma-3)(20\varsigma^2+\varsigma-6)}\tau^4 \\ & +\mathcal{O}(\tau^5)\end{aligned} \tag{8}$$

$$\begin{aligned}H^2 =& \frac{1}{4\varsigma-\varpi^2}\Bigg[\frac{4H_0^2\varsigma}{-\upsilon_2^2}\left(\Omega_{\mathrm{r},0}a^{-4}+\Omega_{\mathrm{m},0}a^{-3}+\Omega_{\Lambda,0}\right)+ \\ & 2H\varpi\dot{\varpi}+\dot{\varpi}^2 \\ & +\frac{4\left[H\varsigma\left(\varpi^2-1\right)+\varsigma\varpi\dot{\varpi}\right]^2}{\varsigma-\varpi^2}\Bigg].\end{aligned}$$

## III METHODOLOGY

Blahblahblahblahblah

## IV BLAH

Methodology text goes here.

***SL: Describe the numerical ODE solver, initial conditions from power-series expansion, rescaling of the scale factor, and comparison with*** ΛCDM***.***

## V RESULTS

Results text goes here.

***SL: Present plots of the expansion history, equation of state, and comparison with*** ΛCDM***.***

## VI DISCUSSION

Discussion text goes here.

## VII CONCLUSION

Conclusion text goes here.

## ACKNOWLEDGMENTS

Acknowledgments go here.

[1] C.-Y. Lin, M. P. Hobson, and A. N. Lasenby, Phys. Rev. D **98**, 064026 (2018), arXiv:1802.02350 [gr-qc].
[2] C.-Y. Lin, M. P. Hobson, and A. N. Lasenby, Phys. Rev. D **101**, 064038 (2020), arXiv:1910.14197 [gr-qc].